Introduction to Derivatives

Assignment 2

1) It is now June. A company knows that it will sell 5,000 barrels of crude oil in September. It uses the October CME Group futures contract to hedge the price it will receive. Each contract is on 1,000 barrels of ''light sweet crude.'' What position should it take? What price risks is it still exposed to after taking the position? 2 pts

Answer:
It should short five contracts. It has basis risk. It is exposed to the difference between the October futures price and the spot price of light sweet crude at the time it closes out its position in September. It is also possibly exposed to the difference between the spot price of light sweet crude and the spot price of the type of oil it is selling.

2) A cattle farmer expects to have 120,000 pounds of live cattle to sell in three months. The live cattle futures contract traded by the CME Group is for the delivery of 40,000 pounds of cattle. How can the farmer use the contract for hedging? From the farmer's viewpoint, what are the pros and cons of hedging?

Answer: The farmer can short 3 contracts that have 3 months to maturity. If the price of cattle falls, the gain on the futures contract will offset the loss on the sale of the cattle. If the price of cattle rises, the gain on the sale of the cattle will be offset by the loss on the futures contract. Using futures contracts to hedge has the advantage that the farmer can greatly reduce the uncertainty about the price that will be received. Its disadvantage is that the farmer no longer gains from favorable movements in cattle prices.

3) A 10-year 8% coupon bond currently sells for $90. A 10-year 4% coupon bond currently sells for $80. The bond coupons are paid every 6 months. What is the 10-year zero rate? (*Hint*: Consider taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds.) Assume continuous compounding in all cases. 3 pts

Answer:

Taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds leads to the following cash flows

$$\text{Year } 0: 90 - 2\times 80 = -70$$

$$\text{Year } 10: 200 - 100 = 100$$

because the coupons cancel out. \$100 in 10 years time is equivalent to \$70 today. The 10-year rate, R, (continuously compounded) is therefore given by

$$100 = 70e^{10R}$$

The rate is

$$\frac{1}{10}\ln\frac{100}{70} = 0.0357$$

or 3.57% per annum.

4) The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of \$4 every 6 months currently sells for \$94.84. A 2year Treasury bond that will pay coupons of \$5 every 6 months currently sells for \$97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year continuously compounded Treasury zero rates. It is given to you that 6m Treasury bill yields are based on semi-annual compounding and 1-year Treasury bills are based on annual compounding. 3 pts

Answer:

The 6-month Treasury bill provides a return of $6/94 = 6.383\%$ in six months. This is $2\times 6.383 = 12.766\%$ per annum with semiannual compounding or $2\ln(1.06383) = 12.38\%$ per annum with continuous compounding. The 12-month rate is $11/89 = 12.360\%$ with annual compounding or $\ln(1.1236) = 11.65\%$ with continuous compounding.

For the $1\frac{1}{2}$ year bond, we must have

$$4e^{-0.1238\times 0.5} + 4e^{-0.1165\times 1} + 104e^{-1.5R} = 94.84$$

where R is the $1\frac{1}{2}$ year zero rate. It follows that

$$3.76 + 3.56 + 104e^{-1.5R} = 94.84$$

$$e^{-1.5R} = 0.8415$$

$$R = 0.115$$

or 11.5%. For the 2-year bond, we must have

$$5e^{-0.1238\times 0.5} + 5e^{-0.1165\times 1} + 5e^{-0.115\times 1.5} + 105e^{-2R} = 97.12$$

where R is the 2-year zero rate. It follows that

$$e^{-2R} = 0.7977$$

$$R = 0.113$$

or 11.3%.